

Supplement S2: Complete Derivations

Mathematical Proofs for All 33 Type I Predictions Brieuc de La Fournière
Independent researcher, Beaune, France July 2026 | v3.5

Complete Mathematical Derivations for All 33 Type I Predictions

This supplement provides mathematical derivations for the 33 Type I dimensionless predictions in the K_7 framework (Relations #1–#33, all Lean-certified). Each derivation proceeds from topological definitions to numerical predictions. Two additional BH remnant topological predictions (Relations #34–#35 in §24.3b) are included as an appendix; these are classified as Type IV structural diagnostics in the main 95-observable dataset.

Status: All 33 Type I observables Lean-certified. 213 certificate conjuncts, 15 classified axioms (of which 4 external data packages, see main §8.1), 0 sorry.

Note on the 95-observable dataset: The main paper presents 95 observables across 4 types (33 I + 19 II + 21 III + 22 IV). This supplement covers the 33 Type I predictions (direct algebraic from topology). Type II extensions (19 one-step physical extractions), Type III dynamical results (21 multi-step chains), and Type IV structural diagnostics (22 internal consistency checks, including BH remnant M_{res} and N_{QNM} from Pinčák et al. 2026, main text ref. [42]) are documented in Supplement S3 and the main text §4–§6.

The topological constants that determine these relations are described in S1.

Table of Contents

- Part I: Foundations
- Part II: Foundational Theorems (4 relations)
- Part III: Gauge Sector (2 relations)
- Part IV: Lepton Sector (3 relations)
- Part V: Quark Sector (1 relation)
- Part VI: Neutrino Sector (4 relations)
- Part VII: Higgs & Cosmology (3 relations)
- Part VIII: Summary Table
- Part IX: Observable Catalog

0. What “Derivation” Means in the K_7 Framework

Before presenting derivations, we clarify the logical structure:

0.1 Inputs vs Outputs

Inputs (taken as given): - The octonion algebra $\mathbb{O}$ and its automorphism group $G_2 = \text{Aut}(\mathbb{O})$ - The $E_8 \times E_8$ gauge structure - The K_7 manifold (compact G_2 7-manifold with $b_2 = 21$, $b_3 = 77$; explicit metric certified in [A]; topological construction via Joyce-Karigiannis Z_2^3 orbifold route (S1 §8.4); explicit closed-form neck ansatz with Wirtinger certificate (S1 §8.4, [D]))

Outputs (derived from inputs): - The 33 Type I dimensionless predictions

0.2 What We Do NOT Claim

- That $\mathbb{O} \rightarrow G_2 \rightarrow K_7$ is the unique geometry for physics
- That the formulas are uniquely determined by geometric principles
- That the selection rule for specific combinations ($b_2/(b_3 + \dim G_2)$ vs b_2/b_3) is understood

0.3 What We Observe

- Given the inputs, the outputs follow by algebra
- The outputs match experiment to 0.73% mean deviation (PDG 2024, NuFIT 6.0, 33 Type I core)
- No continuous parameters are fitted

0.4 Torsion Independence

Important: All 33 Type I predictions use only topological invariants. The torsion T does not appear in any formula. Therefore: - Predictions depend only on topology, not on the actual torsion value - The value $\kappa_T = 1/61$ is a topological bound, not a prediction ingredient

0.5 Results from Analysis of the NK-Certified Metric

Analysis of the NK-certified metric ([B]) computed all invariants directly from the explicit metric. Key results:

- **Topological realization:** The pair $(b_2, b_3) = (21, 77)$ is realized topologically by the Joyce-Karigiannis Z_2^3 orbifold route (S1 §8.4); an explicit closed-form neck ansatz with five-layer Wirtinger certificate is established in [D]. Orthogonal TCS is excluded by parity ($b_2+b_3=98$ even). Any Betti decomposition via Mayer-Vietoris is conditional on the building block identification, which is open.
- $\nu = 0$: Lean-certified in TCSConstruction.lean. For rectangular TCS ($k_+=k_-=1$), $\nu=0$ automatically (CGN invariant); this is conditional on TCS realization.
- **New topological formula:** $V_{\min} = \sqrt{(\text{Vol}(K_7)/11)}$, where $11 = b_3/n = 77/7$. NK numerical: 219.90; formula: 221.24 (0.6% ✓). Identity: $V_{\min}^2 \times b_3/n = \text{Vol}(K_7)$: a universal G_2 relation.

- **$U(1)^2$ exact symmetry:** The T^2 directions θ and ψ are degenerate to 2×10^{-5} in the metric eigenvalues, propagating to $S_\theta = S_\psi = 6.1265$ in all period integrals (exact to 2.6×10^{-8}).
- **SM gauge group origin:** NOT from ADE singularities (TCS is smooth). From $g_2 \subset so(7)$ spectral structure and $so(8) = g_2 \oplus L \oplus R$ triality $\rightarrow N_{\text{gen}} = 3$.

Part I: Foundations

1. Status Classification

Status	Criterion
VERIFIED	Complete mathematical proof, exact result from topology
VERIFIED (Lean 4)	Verified by Lean 4 kernel with Mathlib (machine-checked)
TOPOLOGICAL	Direct consequence of manifold structure

2. Notation

Symbol	Value	Definition
$\dim(E_8)$	248	E_8 Lie algebra dimension
$\text{rank}(E_8)$	8	E_8 Cartan subalgebra dimension
$\dim(G_2)$	14	G_2 holonomy group dimension
$\dim(K_7)$	7	Internal manifold dimension
$b_2(K_7)$	21	Second Betti number
$b_3(K_7)$	77	Third Betti number
H^*	99	Effective cohomology = $b_2 + b_3 + 1$
$\dim(J_3(O))$	27	Exceptional Jordan algebra dimension
N_{gen}	3	Number of fermion generations
p_2	2	Binary duality parameter
Weyl	5	Weyl factor: $(\dim(G_2)+1)/N_{\text{gen}} = b_2/N_{\text{gen}} - p_2 = \dim(G_2) - \text{rank}(E_8) - 1$

Part II: Foundational Theorems

3. Relation #1: Generation Number $N_{\text{gen}} = 3$

Statement: The number of fermion generations is exactly 3.

Classification: VERIFIED (three independent derivations)

Proof Method 1: Fundamental Topological Constraint

Theorem: For G_2 holonomy manifold K_7 with E_8 gauge structure:

$$(\text{rank}(E_8) + N_{\text{gen}}) \cdot b_2(K_7) = N_{\text{gen}} \cdot b_3(K_7)$$

Derivation:

$$(8 + N_{\text{gen}}) \times 21 = N_{\text{gen}} \times 77$$

$$168 + 21 \cdot N_{\text{gen}} = 77 \cdot N_{\text{gen}}$$

$$168 = 56 \cdot N_{\text{gen}}$$

$$N_{\text{gen}} = \frac{168}{56} = 3$$

Verification: - LHS: $(8 + 3) \times 21 = 231$ - RHS: $3 \times 77 = 231$ ✓

Proof Method 2: Atiyah-Singer Index Theorem

$$\text{Index}(D_A) = \left(77 - \frac{8}{3} \times 21\right) \times \frac{1}{7} = 3$$

Status: VERIFIED ■

4. Relation #2: Hierarchy Parameter $\tau = 3472/891$

Statement: The hierarchy parameter is exactly rational.

Classification: VERIFIED

Proof

Step 1: Definition from topological integers

$$\tau := \frac{\dim(E_8 \times E_8) \cdot b_2(K_7)}{\dim(J_3(\mathbb{O})) \cdot H^*}$$

Step 2: Substitute values

$$\tau = \frac{496 \times 21}{27 \times 99} = \frac{10416}{2673}$$

Step 3: Reduce

$$\text{gcd}(10416, 2673) = 3$$

$$\tau = \frac{3472}{891}$$

Step 4: Prime factorization

$$\tau = \frac{2^4 \times 7 \times 31}{3^4 \times 11}$$

Step 5: Numerical value

$$\tau = 3.8967452300785634...$$

Status: VERIFIED ■

5. Relation #3: Torsion Capacity $\kappa_T = 1/61$

Statement: The topological torsion capacity equals exactly 1/61.

Classification: TOPOLOGICAL (structural parameter, not physical prediction)

Proof

Step 1: Define from cohomology

$$61 = b_3(K_7) - \dim(G_2) - p_2 = 77 - 14 - 2 = 61$$

Step 2: Formula

$$\kappa_T = \frac{1}{b_3 - \dim(G_2) - p_2} = \frac{1}{61}$$

Step 3: Geometric interpretation - 61 = effective degrees of freedom available for torsional deformation - $61 = \dim(F_4) + N_gen^2 = 52 + 9$

Clarification

Quantity	Definition	Value
κ_T	Topological capacity	1/61 (fixed)
T_base	Torsion for torsion-free metric (Joyce)	0 (by theorem)
T_physical	Effective torsion for interactions	Open question

Role in predictions: κ_T appears in only one formula (α^{-1} , as a small correction term $\det(g) \times \kappa_T \approx 0.033$). The other 32 Type I predictions are independent of torsion capacity. It is primarily a structural parameter characterizing K_7 , not a directly measured observable.

Joyce’s theorem: For compact G_2 manifolds satisfying perturbation bounds, guarantees existence of a torsion-free metric. The Newton-Kantorovich certificate ([A]) establishes these bounds for K_7 .

Status: TOPOLOGICAL (structural, not predictive) ■

6. Relation #4: Metric Determinant $\det(g) = 65/32$

Statement: The K_7 metric determinant is exactly $65/32$.

Classification: METRIC NORMALIZATION (not TOPOLOGICAL, see S1 §10.3)

Proof

Step 1: Define from structural normalization integers

$$\det(g) = p_2 + \frac{1}{b_2 + \dim(G_2) - N_{gen}}$$

Step 2: Compute denominator

$$b_2 + \dim(G_2) - N_{gen} = 21 + 14 - 3 = 32$$

Step 3: Compute determinant

$$\det(g) = 2 + \frac{1}{32} = \frac{65}{32}$$

Step 4: Alternative derivation

$$\det(g) = \frac{\text{Weyl} \times (\text{rank}(E_8) + \text{Weyl})}{2^5} = \frac{5 \times 13}{32} = \frac{65}{32}$$

Verification: The analytical metric $g = (65/32)^{1/7} \times I_7$ has $\det(g) = [(65/32)^{1/7}]^7 = 65/32$ exactly, consistent with the topological formula.

Epistemic note: Three independent algebraic paths converge to $65/32$, which is suggestive but does not constitute a derivation from topology: the metric optimization was constrained to this normalization target, and the formulas were identified post-hoc. See S1 §10.3 for full discussion. Six observables depending on $\det(g)_{\text{num}}$ or $\det(g)_{\text{den}}$ carry this normalization dependence.

Status: STRUCTURAL (metric normalization, algebraically exact; see S1 §10.3) ■

Part III: Gauge Sector

7. Relation #5: Weinberg Angle $\sin^2\theta_W = 3/13$

Statement: The weak mixing angle has exact rational form $3/13$.

Classification: VERIFIED

Proof

Step 1: Define ratio from Betti numbers

$$\sin^2 \theta_W = \frac{b_2(K_7)}{b_3(K_7) + \dim(G_2)} = \frac{21}{77 + 14} = \frac{21}{91}$$

Step 2: Simplify

$$\begin{aligned} \gcd(21, 91) &= 7 \\ \sin^2 \theta_W &= \frac{3}{13} = 0.230769... \end{aligned}$$

Step 3: Experimental comparison

Quantity	Value
Experimental (PDG 2024)	0.23122 ± 0.00004
Prediction	0.230769
Deviation	0.195%

Status: VERIFIED ■

8. Relation #6: Strong Coupling $\alpha_s = \sqrt{2}/12$

Statement: The strong coupling at M_Z scale.

Classification: TOPOLOGICAL

Proof

Formula:

$$\alpha_s(M_Z) = \frac{\sqrt{2}}{\dim(G_2) - p_2} = \frac{\sqrt{2}}{14 - 2} = \frac{\sqrt{2}}{12}$$

Components: - $\sqrt{2}$: E_8 root length - 12 = $\dim(G_2)$ - p_2 : Effective gauge degrees of freedom

Numerical value: $\alpha_s = 0.117851$

Experimental comparison:

Quantity	Value
Experimental	0.1180 ± 0.0009
Prediction	0.11785
Deviation	0.127%

Status: TOPOLOGICAL ■

Part IV: Lepton Sector

9. Relation #7: Koide Parameter $Q = 2/3$ **Statement:** The Koide parameter equals exactly $2/3$.**Classification:** VERIFIED**Proof***Formula:*

$$Q_{\text{Koide}} = \frac{\dim(G_2)}{b_2(K_7)} = \frac{14}{21} = \frac{2}{3}$$

Physical definition:

$$Q = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2}$$

Experimental comparison:

Quantity	Value
Experimental	0.666661 ± 0.000007
Prediction	0.666667
Deviation	0.0009%

Status: VERIFIED ■**10. Relation #8: Tau-Electron Mass Ratio $m_\tau/m_e = 3477$** **Statement:** The tau-electron mass ratio is exactly 3477.**Classification:** VERIFIED**Proof***Formula:*

$$\begin{aligned} \frac{m_\tau}{m_e} &= \dim(K_7) + 10 \cdot \dim(E_8) + 10 \cdot H^* \\ &= 7 + 10 \times 248 + 10 \times 99 = 7 + 2480 + 990 = 3477 \end{aligned}$$

Prime factorization:

$$3477 = 3 \times 19 \times 61 = N_{\text{gen}} \times \text{prime}(8) \times \kappa_T^{-1}$$

Period integral cross-confirmation: Analysis of the NK-certified metric ([B], §5) extracts associative 3-cycle volumes directly. The SD associative volumes yield $\ln(m_\tau/m_e) = 8.154$ from the Fano-plane volume hierarchy, giving $e^{8.154} = 3477 \checkmark$. This provides an independent geometric confirmation of the algebraic prediction.

Experimental comparison:

Quantity	Value
Experimental	3477.15 ± 0.05
Prediction	3477 (exact)
Deviation	0.0043%

Status: VERIFIED ■

11. Relation #9: Muon-Electron Mass Ratio

Statement: $m_\mu/m_e = 27^\varphi$

Classification: TOPOLOGICAL

Proof

Formula:

$$\frac{m_\mu}{m_e} = [\dim(J_3(\mathbb{O}))]^\varphi = 27^\varphi = 207.012$$

Components: - $27 = \dim(J_3(\mathbb{O}))$: Exceptional Jordan algebra - $\varphi = (1+\sqrt{5})/2$: Golden ratio from McKay correspondence $E_8 \leftrightarrow 2I$ (binary icosahedral group)

Experimental comparison:

Quantity	Value
Experimental	206.768
Prediction	207.01
Deviation	0.1179%

Note on φ : The golden ratio is the only non-integer constant among the 33 Type I predictions and does not appear in the 20 structural constants (S3.3). Its derivation from E_8 via the McKay correspondence is well-established mathematically but constitutes an additional identification step beyond the direct topological reasoning used for the other 32 predictions.

Status: TOPOLOGICAL (with caveat: φ external to the 20 structural constants) ■

Part V: Quark Sector

12. Relation #10: Strange-Down Ratio $m_s/m_d = 20$

Statement: The strange-down quark mass ratio is exactly 20.

Classification: VERIFIED

Proof

Formula:

$$\frac{m_s}{m_d} = p_2^2 \times \text{Weyl} = 4 \times 5 = 20$$

Geometric interpretation: - $p_2^2 = 4$: Binary structure squared - Weyl = 5: Pentagonal symmetry

Experimental comparison:

Quantity	Value
Experimental	20.0 ± 1.0
Prediction	20 (exact)
Deviation	0.00%

Status: VERIFIED ■

12b. Relation #10b: Charm-Strange Ratio $m_c/m_s = 246/21$

Statement: The charm-strange quark mass ratio.

Classification: TOPOLOGICAL

Proof

Formula:

$$\frac{m_c}{m_s} = \frac{\dim(E_8) - p_2}{b_2(K_7)} = \frac{248 - 2}{21} = \frac{246}{21} = 11.714...$$

Components: - $246 = \dim(E_8) - p_2$: Effective E_8 dimension - $21 = b_2(K_7)$: Second Betti number

Experimental comparison:

Quantity	Value
Experimental	11.7 ± 0.3
Prediction	11.714

Quantity	Value
Deviation	0.12%

TCS-level alternative: The charm-strange ratio also admits a decomposition at the TCS building-block level:

$$\frac{m_c}{m_s} = b_2(M_1) + \frac{\dim(K_7)}{b_2(M_2)} = 11 + \frac{7}{10} = \frac{117}{10} = 11.700$$

This additive form reflects the adiabatic decomposition of the KK mass operator across the two TCS halves. The M_1 contribution ($b_2(M_1) = 11$) dominates, with a geometric correction from M_2 ($\dim(K_7)/b_2(M_2) = 7/10$). Both formulas are within 0.05σ of the experimental central value.

Status: TOPOLOGICAL ■

12c. Relation #10c: Bottom-Top Ratio $m_b/m_t = 1/42$

Statement: The bottom-top quark mass ratio involves the constant $42 = p_2 \times N_{\text{gen}} \times \dim(K_7)$.

Classification: TOPOLOGICAL

Proof

Step 1: Define the structural constant

$$42 = p_2 \times N_{\text{gen}} \times \dim(K_7) = 2 \times 3 \times 7$$

This constant 42 also equals $2 \times b_2 = 2 \times 21$.

Step 2: Formula

$$\frac{m_b}{m_t} = \frac{b_0}{42} = \frac{1}{42} = 0.02381\dots$$

Components: - $b_0 = 1$: Zeroth Betti number - 42: Structural constant from K_7 geometry

Experimental comparison:

Quantity	Value
Experimental	0.024 ± 0.001
Prediction	0.02381
Deviation	0.79%

Geometric interpretation: The same constant 42 appears in the cosmological ratio $\Omega_{\text{DM}}/\Omega_b = (1 + 42)/8 = 43/8$ (Section 18b), connecting quark physics to cosmological structure through the K_7 geometry.

Status: TOPOLOGICAL ■

12d. Relation #10d: Up-Down Ratio $m_u/m_d = 79/168$

Statement: The up-down quark mass ratio.

Classification: TOPOLOGICAL

Proof

Formula:

$$\frac{m_u}{m_d} = \frac{b_0 + \dim(E_6)}{|PSL_2(7)|} = \frac{1 + 78}{168} = \frac{79}{168} = 0.4702...$$

Components: - $\dim(E_6) = 78$: Exceptional Lie algebra dimension - $|PSL_2(7)| = 168$: Order of the simple group $PSL_2(7) = \text{rank}(E_8) \times b_2$

Experimental comparison:

Quantity	Value
Experimental	0.47 ± 0.03
Prediction	0.4702
Deviation	0.05%

Status: TOPOLOGICAL ■

Part V-B: CKM Matrix

12e. Relation #10e: Cabibbo Angle $\sin^2\theta_{12}(\text{CKM}) = 7/31$

Statement: The CKM Cabibbo mixing angle.

Classification: TOPOLOGICAL

Proof

Formula:

$$\sin^2\theta_{12}^{CKM} = \frac{\dim(\text{fund}_{E_7})}{\dim(E_8)} = \frac{56}{248} = \frac{7}{31} = 0.2258...$$

Alternative expressions: - $(b_3 - b_2)/\dim(E_8) = (77 - 21)/248 = 56/248$ - $(2b_2 + \dim(G_2))/\dim(E_8) = (42 + 14)/248 = 56/248$

Experimental comparison:

Quantity	Value
Experimental	0.2250 ± 0.0006
Prediction	0.2258
Deviation	0.36%

Status: TOPOLOGICAL ■

12f. Relation #10f: Wolfenstein A Parameter = 83/99

Statement: The Wolfenstein A parameter of the CKM matrix.

Classification: TOPOLOGICAL

Proof

Formula:

$$A_{\text{Wolf}} = \frac{\text{Weyl} + \dim(E_6)}{H^*} = \frac{5 + 78}{99} = \frac{83}{99} = 0.8384...$$

Alternative expression: - $(b_3 + p_2 \times N_{\text{gen}})/H^* = (77 + 6)/99 = 83/99$

Experimental comparison:

Quantity	Value
Experimental	0.836 ± 0.015
Prediction	0.8384
Deviation	0.29%

Status: TOPOLOGICAL ■

12g. Relation #10g: CKM θ_{23} Mixing $\sin^2\theta_{23}(\text{CKM}) = 1/24$

Statement: The CKM 23-mixing angle.

Classification: TOPOLOGICAL

Proof*Formula:*

$$\sin^2 \theta_{23}^{CKM} = \frac{\dim(K_7)}{|PSL_2(7)|} = \frac{7}{168} = \frac{1}{24} = 0.04167\dots$$

Experimental comparison:

Quantity	Value
Experimental	0.0412 ± 0.0008
Prediction	0.04167
Deviation	1.13%

Status: TOPOLOGICAL ■

Part VI: Neutrino Sector

13. Relation #11: CP Violation Phase $\delta_{\text{CP}} = 197^\circ$ **Statement:** The CP violation phase is exactly 197° .**Classification:** VERIFIED (experimental drift across NuFIT releases discussed in Appendix F; against NuFIT 6.1 the prediction sits inside the 1σ band)**Proof***Formula:*

$$\delta_{CP} = \dim(K_7) \cdot \dim(G_2) + H^* = 7 \times 14 + 99 = 98 + 99 = 197^\circ$$

Experimental comparison:

Quantity	Value
Experimental (NuFIT 5.2, 2022)	$197^\circ \pm 24^\circ$
Experimental (NuFIT 6.0, Oct 2024)	$177^\circ \pm 20^\circ$
Experimental (NuFIT 6.1, 2025, NO w/o SK-atm)	$207^\circ +23/-20$
Prediction	197° (exact)
Deviation (vs NuFIT 5.2)	0.00%
Deviation (vs NuFIT 6.0)	11.3% (edge of 1σ)
Deviation (vs NuFIT 6.1)	4.8% (inside 1σ)

Experimental drift: The central value shifted from 197° to 177° between NuFIT 5.2 and 6.0, then to 207° in NuFIT 6.1 (2025, driven mainly by the T2K+NOvA joint analysis); the NuFIT collaboration flags

δ_{CP} as one of the least constrained parameters, with substantial dependence on reactor flux assumptions. The two most recent releases bracket the prediction on both sides, and against NuFIT 6.1 the value 197° sits inside the 1σ band. DUNE (first beam targeted 2031; physics run late 2030s–2040s) will resolve this definitively.

The canonical framework prediction remains $\delta_{\text{CP}} = 197^\circ$. A potential compactification correction (62/69 factor, yielding 177.01°) is documented in Appendix F as contingent on experimental confirmation. See also §4.2 of the main paper.

Status: VERIFIED, original prediction, pending DUNE ■

14. Relation #12: Reactor Mixing Angle $\theta_{13} = \pi/21$

Statement: The reactor neutrino mixing angle.

Classification: TOPOLOGICAL

Proof

Formula:

$$\theta_{13} = \frac{\pi}{b_2(K_7)} = \frac{\pi}{21} = 8.571^\circ$$

Experimental comparison:

Quantity	Value
Experimental (NuFIT 6.0)	$8.54^\circ \pm 0.12^\circ$
Prediction	8.571°
Deviation	0.37%

TCS-level refinement: At the building-block level, the reactor angle admits a purely rational formula:

$$\theta_{13} = \frac{\dim(E_8) - \dim(G_2) \cdot b_2(M_1)}{b_2(M_1)} = \frac{248 - 14 \times 11}{11} = \frac{94}{11} = 8.545^\circ$$

The numerator $94 = \dim(E_8) - \dim(G_2) \times b_2(M_1)$ is the residual E_8 gauge content after subtracting the G_2 holonomy contribution to M_1 . This formula is $5.8\times$ more precise (0.064% vs 0.37%) and uses only algebraic invariants (no π), but works in degrees rather than radians.

Status: TOPOLOGICAL ■

15. Relation #13: Atmospheric Mixing Angle θ_{23}

Statement: The atmospheric neutrino mixing angle.

Classification: TOPOLOGICAL

Proof

Formula:

$$\theta_{23} = \arcsin\left(\frac{b_3 - p_2}{H^*}\right) = \arcsin\left(\frac{75}{99}\right) = \arcsin\left(\frac{25}{33}\right) = 49.251^\circ$$

Components: - $b_3 = 77$: Third Betti number (3-cycles of K_7) - $p_2 = 2$: Pontryagin class contribution (spin structure correction) - $H^* = 99$: Effective cohomology ($b_2 + b_3 + 1$)

Physical interpretation: The atmospheric mixing angle θ_{23} governs τ - μ flavor mixing. The formula $(b_3 - p_2)/H^*$ represents the relative weight of spin-corrected 3-cycles in the total cohomology. This captures how the τ - μ sector couples through the 3-cycle topology of K_7 , with the Pontryagin correction accounting for the spin structure that distinguishes fermionic generations.

Experimental comparison:

Quantity	Value
Experimental (NuFIT 5.3)	$49.3^\circ \pm 1.0^\circ$
Prediction	49.251°
Deviation	0.10%

Status: TOPOLOGICAL ■

16. Relation #14: Solar Mixing Angle θ_{12}

Statement: The solar neutrino mixing angle.

Classification: TOPOLOGICAL

Proof

Formula:

$$\theta_{12} = \arctan\left(\sqrt{\frac{\delta}{\gamma_{\text{GIFT}}}}\right) = 33.419^\circ$$

Components: - $\delta = 2\pi/\text{Weyl}^2 = 2\pi/25$ - $\gamma_{\text{GIFT}} = 511/884$ (historical symbol name)

Derivation of γ_{GIFT} :

$$\gamma_{\text{GIFT}} = \frac{2 \cdot \text{rank}(E_8) + 5 \cdot H^*}{10 \cdot \dim(G_2) + 3 \cdot \dim(E_8)} = \frac{511}{884}$$

Experimental comparison:

Quantity	Value
Experimental (NuFIT 5.3)	$33.41^\circ \pm 0.75^\circ$
Prediction	33.40°
Deviation	0.030%

Status: TOPOLOGICAL ■

16b. PMNS Matrix: $\sin^2$ Form

The PMNS mixing angles can also be expressed directly as $\sin^2$ values, providing alternative topological formulas.

Relation #14b: $\sin^2\theta_{12}(\text{PMNS}) = 4/13$

Formula:

$$\sin^2 \theta_{12}^{PMNS} = \frac{b_0 + N_{gen}}{\alpha_{sum}} = \frac{1 + 3}{13} = \frac{4}{13} = 0.3077...$$

Components: - $\alpha_sum = 13$: Anomaly coefficient sum - $b_0 + N_gen = 4$: Cohomological + generation count

Quantity	Value
Experimental	0.307 ± 0.013
Prediction	0.3077
Deviation	0.23%

Relation #14c: $\sin^2\theta_{23}(\text{PMNS}) = 6/11$

Formula:

$$\sin^2 \theta_{23}^{PMNS} = \frac{D_{bulk} - \text{Weyl}}{D_{bulk}} = \frac{11 - 5}{11} = \frac{6}{11} = 0.5455...$$

Alternative expression: - $42/b_3 = 42/77 = 6/11$ (after reduction)

Quantity	Value
Experimental	0.546 ± 0.021
Prediction	0.5455
Deviation	0.10%

Relation #14d: $\sin^2\theta_{13}(\text{PMNS}) = 11/496$ **Formula:**

$$\sin^2\theta_{13}^{PMNS} = \frac{D_{bulk}}{\dim(E_8 \times E_8)} = \frac{11}{496} = 0.02218\dots$$

Quantity	Value
Experimental	0.0220 ± 0.0007
Prediction	0.02218
Deviation	0.81%

Status: TOPOLOGICAL ■

Part VII: Higgs & Cosmology

17. Relation #15: Higgs Coupling $\lambda_H = \sqrt{17/32}$ **Statement:** The Higgs quartic coupling is determined by G_2 holonomy parameters.**Classification:** STRUCTURAL (uses the imposed metric normalization $\det(g) = 65/32$; see main paper §2.3)**Proof***Formula:*

$$\lambda_H = \frac{\sqrt{\dim(G_2) + N_{gen}}}{\det(g)_{den}} = \frac{\sqrt{14+3}}{32} = \frac{\sqrt{17}}{32} = 0.12885$$

Components: - $\dim(G_2) + N_{gen} = 17$: holonomy dimension plus generation count - $\det(g)_{den} = 32$: metric determinant denominator (= $b_2 + \dim(G_2) - N_{gen}$)*Physical interpretation:* The Higgs self-coupling combines the holonomy group dimension with the generation count, normalized by the metric determinant scale. The square root reflects the quadratic nature of the Higgs potential.*Numerical value:* $\lambda_H = \sqrt{17/32} = 0.128847\dots$ *Experimental comparison:*

Quantity	Value
Experimental (PDG 2024)	0.129 ± 0.001
Prediction	$\sqrt{17/32} = 0.12885$
Deviation	0.12%

TCS Alternative

A purely rational TCS formula is also available:

$$\lambda_H^{TCS} = \frac{b_2(M_1)}{b_3 + b_2(M_2)} = \frac{11}{87} = 0.12644$$

where $b_2(M_1) = 11$ and $b_2(M_2) = 10$ are the Betti numbers of the two TCS building blocks. This formula is lower-complexity (cost 3 vs cost 6) and connects directly to the twisted connected sum decomposition. Against a running coupling value $\lambda_H(\mu_{\text{high}}) \approx 0.126$, it achieves 0.35% deviation, though against the PDG 2024 pole value (0.129) it gives 1.98%.

Status: STRUCTURAL (metric-normalization dependent) ■

17b. Boson Mass Ratios

Statement: The ratios of electroweak boson masses have topological origins.

Classification: VERIFIED

Relation: $m_W/m_Z = 37/42$

Formula:

$$\frac{m_W}{m_Z} = \frac{2b_2 - \text{Weyl}}{2b_2} = \frac{42 - 5}{42} = \frac{37}{42}$$

Physical interpretation: - $2b_2 = 42$ is the structural constant ($= p_2 \times b_2$) - Weyl = 5 is the triple identity factor - The ratio involves (structural_const - Weyl) / structural_const

Note: The true Euler characteristic $\chi(K_7) = 0$ for odd-dimensional manifolds. The constant $42 = 2b_2$ is a distinct topological invariant.

Numerical value: $m_W/m_Z = 0.8810$

Experimental comparison:

Quantity	Value
Experimental	0.8815 ± 0.0002
Prediction	0.8810
Deviation	0.06%

Relation: $m_H/m_t = 56/77$

Formula:

$$\frac{m_H}{m_t} = \frac{fund(E_7)}{b_3} = \frac{56}{77} = \frac{8}{11}$$

Numerical value: $m_H/m_t = 0.7273$

Quantity	Value
Experimental	0.725 ± 0.003
Prediction	0.7273
Deviation	0.31%

Relation: $m_H/m_W = 81/52$

Formula:

$$\frac{m_H}{m_W} = \frac{N_{gen} + \dim(E_6)}{\dim(F_4)} = \frac{3 + 78}{52} = \frac{81}{52}$$

Numerical value: $m_H/m_W = 1.5577$

Quantity	Value
Experimental	1.558 ± 0.002
Prediction	1.5577
Deviation	0.02%

Status: VERIFIED ■

18. Relation #16: Dark Energy Density $\Omega_{DE} = \ln(2) \times 98/99$

Statement: The dark energy density fraction emerges from information-theoretic structure of K_7 moduli.

Classification: TOPOLOGICAL

Proof

Formula:

$$\Omega_{DE} = \ln(2) \times \frac{b_2 + b_3}{H^*} = \ln(2) \times \frac{98}{99} = 0.6861$$

Components: - $\ln(2)$: binary entropy factor - $b_2 + b_3 = 98$: total non-trivial cohomological content - $H^* = 99 = b_2 + b_3 + 1$: full cohomological dimension

Physical interpretation: The dark energy fraction combines a binary information factor with the ratio of non-trivial to total cohomological content. The $\ln(2)$ reflects the fundamental binary structure of the moduli space partition.

Numerical value: $\Omega_{DE} = \ln(2) \times 98/99 = 0.68615\dots$

Experimental comparison:

Quantity	Value
Experimental (Planck PR4)	0.6847 ± 0.005
Prediction	$\ln(2) \times 98/99 = 0.6861$
Deviation	0.21%

TCS Alternative

A purely rational TCS formula is also available:

$$\Omega_{DE}^{TCS} = 1 - \frac{\chi(K3)}{b_3} = 1 - \frac{24}{77} = \frac{53}{77} = 0.68831$$

where $\chi(K3) = 24$ is the Euler characteristic of the K3 fiber. This formula is lower-complexity (rational, cost 2 vs cost 6), and the interpretation is transparent: $\Omega_{\text{matter}} = \chi(K3)/b_3$ counts the K3-fiber fraction of the 3-cycle moduli, with the rest being vacuum energy. Against Planck 2018 ($\Omega_{\Lambda} = 0.6889$), it achieves 0.085% deviation, though against Planck PR4 (0.6847) it gives 0.53%.

Status: TOPOLOGICAL ■

19. Relation #17: Spectral Index n_s

Statement: The primordial scalar spectral index.

Classification: VERIFIED

Proof

Formula:

$$n_s = \frac{\zeta(D_{bulk})}{\zeta(Weyl)} = \frac{\zeta(11)}{\zeta(5)} = 0.9649$$

Components: - $\zeta(11)$: From 11D bulk spacetime - $\zeta(5)$: From Weyl factor

Experimental comparison:

Quantity	Value
Experimental (Planck 2020)	0.9649 ± 0.0042
Prediction	0.9649
Deviation	0.004%

Status: VERIFIED ■

19b. Relation #17c: Dark Matter to Baryon Ratio $\Omega_{DM}/\Omega_b = 43/8$

Statement: The dark matter to baryon density ratio.

Classification: TOPOLOGICAL

Proof

Formula:

$$\frac{\Omega_{DM}}{\Omega_b} = \frac{b_0 + 42}{\text{rank}(E_8)} = \frac{1 + 42}{8} = \frac{43}{8} = 5.375$$

Components: - $42 = p_2 \times N_{\text{gen}} \times \dim(K_7)$: The same constant appearing in $m_b/m_t = 1/42$ - $\text{rank}(E_8) = 8$: Cartan subalgebra dimension

Experimental comparison:

Quantity	Value
Experimental (Planck 2020)	5.375 ± 0.05
Prediction	5.375
Deviation	0.00%

Status: TOPOLOGICAL ■

19c. Relation #17d: Reduced Hubble Parameter $h = 167/248$

Statement: The reduced Hubble parameter $H_0 = 100h \text{ km/s/Mpc}$.

Classification: TOPOLOGICAL

Proof

Formula:

$$h = \frac{|PSL_2(7)| - b_0}{\dim(E_8)} = \frac{168 - 1}{248} = \frac{167}{248} = 0.6734...$$

Experimental comparison:

Quantity	Value
Experimental (Planck 2020)	0.674 ± 0.005
Prediction	0.6734
Deviation	0.09%

Status: TOPOLOGICAL ■

19d. Relation #17e: Baryon Fraction $\Omega_b/\Omega_m = 5/32$

Statement: The baryon fraction of total matter.

Classification: STRUCTURAL (uses the imposed metric normalization $\det(g) = 65/32$; see main paper §2.3)

Proof

Formula:

$$\frac{\Omega_b}{\Omega_m} = \frac{\text{Weyl}}{\det(g)_{den}} = \frac{5}{32} = 0.15625$$

Experimental comparison:

Quantity	Value
Experimental (Planck 2020)	0.156 ± 0.003
Prediction	0.15625
Deviation	0.16%

Status: STRUCTURAL (metric-normalization dependent) ■

19e. Relation #17f: Amplitude of Fluctuations $\sigma_8 = 17/21$

Statement: The amplitude of matter fluctuations at $8 \text{ h}^{-1} \text{ Mpc}$.

Classification: STRUCTURAL (uses the imposed metric normalization $\det(g) = 65/32$; see main paper §2.3)

Proof

Formula:

$$\sigma_8 = \frac{p_2 + \det(g)_{den}}{42} = \frac{2 + 32}{42} = \frac{34}{42} = \frac{17}{21} = 0.8095\dots$$

Experimental comparison:

Quantity	Value
Experimental (Planck 2020)	0.811 ± 0.006
Prediction	0.8095
Deviation	0.18%

Quantity	Value
----------	-------

Status: STRUCTURAL (metric-normalization dependent) ■

19f. Relation #17g: Primordial Helium Fraction $Y_p = 15/61$

Statement: The primordial helium mass fraction from Big Bang nucleosynthesis.

Classification: TOPOLOGICAL

Proof

Formula:

$$Y_p = \frac{b_0 + \dim(G_2)}{\kappa_T^{-1}} = \frac{1 + 14}{61} = \frac{15}{61} = 0.2459...$$

Experimental comparison:

Quantity	Value
Experimental	0.245 ± 0.003
Prediction	0.2459
Deviation	0.37%

Status: TOPOLOGICAL ■

20. Relation #17b: Matter Density Ω_m

Statement: The matter density fraction derives from dark energy via $\sqrt{\text{Weyl}}$.

Classification: DERIVED (from Weyl triple identity + Ω_{DE})

Proof

Step 1: Establish $\sqrt{\text{Weyl}}$ as structural

From the Weyl Triple Identity (S1, Section 2.3):

$$\text{Weyl} = \frac{\dim(G_2) + 1}{N_{gen}} = \frac{b_2}{N_{gen}} - p_2 = \dim(G_2) - \text{rank}(E_8) - 1 = 5$$

Therefore $\sqrt{\text{Weyl}} = \sqrt{5}$ is a derived quantity.

Step 2: Matter-dark energy ratio

The cosmological density ratio:

$$\frac{\Omega_{DE}}{\Omega_m} = \sqrt{\text{Weyl}} = \sqrt{5}$$

Step 3: Compute Ω_m

Using $\Omega_{DE} = \ln(2) \times (b_2 + b_3)/H^* = 0.6861$ (Relation #16):

$$\Omega_m = \frac{\Omega_{DE}}{\sqrt{\text{Weyl}}} = \frac{\ln(2) \times 98/99}{\sqrt{5}} = \frac{0.6861}{2.236} = 0.3068$$

Step 4: Verify closure

$$\Omega_{total} = \Omega_{DE} + \Omega_m = 0.6861 + 0.3068 = 0.9929 \approx 1$$

Consistent with flat universe ($\Omega_{total} = 1$).

Experimental comparison:

Quantity	Value
Experimental (Planck 2020)	0.3153 ± 0.007
Prediction	0.3068
Deviation	2.7%

Interpretation

The $\sqrt{5}$ ratio between dark energy and matter densities emerges from the same structural constant (Weyl = 5) that determines: - $\det(g) = 65/32$ (metric determinant) - $|\text{W}(E_8)|$ factorization (group theory) - N_{gen}^3 coefficient in $|\text{W}(E_8)|$ (topology)

Status: DERIVED (structural, 2.7% deviation) ■

20b. Relation #18: Fine Structure Constant α^{-1}

Statement: The inverse fine structure constant.

Classification: STRUCTURAL (uses the imposed metric normalization $\det(g) = 65/32$; see main paper §2.3)

Proof

Formula:

$$\alpha^{-1}(M_Z) = \frac{\dim(E_8) + \text{rank}(E_8)}{2} + \frac{H^*}{D_{bulk}} + \det(g) \cdot \kappa_T$$

$$= 128 + 9 + \frac{65}{32} \times \frac{1}{61} = 137.033$$

Components: - 128 = (248 + 8)/2: Algebraic - 9 = 99/11: Bulk impedance - 65/1952: Torsional correction

Experimental comparison:

Quantity	Value
Experimental	137.035999
Prediction	137.033
Deviation	0.002%

Status: STRUCTURAL (metric-normalization dependent) ■

Part VIII: Summary Table

21. The 33 Type I Dimensionless Relations (+ 2 Type IV BH Appendix)

Note: All predictions use only topological invariants (b_2 , b_3 , $\dim(G_2)$, etc.). None depend on the realized torsion value T . Rows labelled STRUCTURAL additionally use the imposed metric normalization $\det(g) = 65/32$ (main §2.3). Relation #19 (Ω_m) is DERIVED from Ω_{DE} via the Weyl triple identity.

#	Relation	Formula	Value	Exp.	Dev.	Status
1	N_gen	Atiyah-Singer	3	3	exact	VERIFIED
2	τ	$496 \times 21 / (27 \times 99)$	$3472/891$	-	-	VERIFIED
3	κ_T	$1/(77-14-2)$	$1/61$	-	-	LEDGER PARAM.*
4	$\det(g)$	$5 \times 13/32$	$65/32$	-	-	STRUCTURAL (norm. target)
5	$\sin^2 \theta_W$	$21/91$	$3/13$	0.23122	0.195%	VERIFIED
6	α_s	$\sqrt{2}/12$	0.11785	0.1180	0.127%	TOPOLOGICAL
7	Q_Koide	$14/21$	$2/3$	0.666661	0.0009%	VERIFIED
8	m_τ/m_e	$7+2480+990$	3477	3477.15	0.0043%	VERIFIED
9	m_μ/m_e	$27^\wedge \varphi$	207.01	206.768	0.118%	TOPOLOGICAL
10	m_s/m_d	4×5	20	20.0	0.00%	VERIFIED
11	δ_{CP}	$7 \times 14 + 99$	197°	$177^\circ \pm 20^{*\circ}$	11.3% (1σ)	VERIFIED
12	θ_{13}	$\pi/21$ (TCS: $94/11$)	8.57°	8.54°	0.37%	TOPOLOGICAL
13	θ_{23}	$\arcsin((b_3-$ $p_2)/H^*)$	49.25°	49.3°	0.10%	TOPOLOGICAL
14	θ_{12}	$\arctan(\dots)$	33.40°	33.41°	0.030%	TOPOLOGICAL

#	Relation	Formula	Value	Exp.	Dev.	Status
15	λ_H	$\sqrt{(\dim(G_2)+N_{\text{gen}})/\det_den}$	$\sqrt{17/32}$	0.129	0.12%	STRUCTURAL
16	Ω_DE	$\ln(2) \times (b_2+b_3)/H^*$	0.6861	0.6847	0.21%	TOPOLOGICAL
17	n_s	$\zeta(11)/\zeta(5)$	0.9649	0.9649	0.004%	VERIFIED
18	α^{-1}	$128+9+\text{corr}$	137.033	137.036	0.002%	STRUCTURAL
19	Ω_m	$\Omega_DE/\sqrt{Weyl}$	0.3068	0.3153	2.7%	DERIVED

*NuFIT 6.0 (Oct 2024); $\pm 20^\circ$ uncertainty; frozen dataset for the deviation statistics of this supplement. The 197° prediction matched NuFIT 5.2 (2022) exactly, and sits inside the NuFIT 6.1 (2025) 1σ band ($207^\circ +23/-20$, deviation 4.8%). See §4.2 of main paper and Appendix F.

* κ_T is a ledger parameter (capacity), not a physical prediction; it does not appear in other formulas. Labelled LEDGER PARAM. to avoid collision with STRUCTURAL, reserved for $\det(g)$ -normalization dependence.

22. Deviation Statistics

Range	Count	Percentage
0.00% (exact)	1	3%
<0.01%	4	12%
0.01-0.1%	4	12%
0.1-0.5%	16	48%
0.5-1%	3	9%
1-5%	4	12%
>5% (δ_CP vs frozen 6.0)	1	3%

Mean deviation: 0.73% (PDG 2024, NuFIT 6.0 frozen dataset; 33 Type I observables with experimental comparison, canonical $\delta_CP=197^\circ$). For the full 95-observable dataset (66 with experimental comparison), see Supplement S3.

23. Statistical Uniqueness of ($b_2=21$, $b_3=77$)

A critical question for any unified framework is whether the specific topological parameters represent overfitting. We conducted comprehensive Monte Carlo validation to address this concern.

Methodology

- **Betti variations:** 100,000 random (b_2 , b_3) configurations
- **Gauge group comparison:** $E_8 \times E_8$, $E_7 \times E_8$, $E_6 \times E_8$, $E_7 \times E_7$, $SO(32)$, $SU(5) \times SU(5)$, etc.
- **Holonomy comparison:** G_2 , $Spin(7)$, $SU(3)$, $SU(4)$
- **Full combinatorial:** 91,896 configurations varying all parameters
- **Local sensitivity:** ± 10 grid around ($b_2=21$, $b_3=77$)

Results

Metric	Value
Total configurations tested	192,349
Configurations better than the K_7 framework	0
K_7 framework mean deviation	0.73% (33 observables, frozen dataset)
Alternative mean deviation	32.9%
P-value	$< 5 \times 10^{-6}$
Significance	$> 4.5\sigma$

Gauge Group Ranking

Rank	Group	Mean Dev.
1	$E_8 \times E_8$	0.73%
2	$E_7 \times E_8$	8.80%
3	$E_6 \times E_8$	15.50%

$E_8 \times E_8$ achieves approximately 12x better agreement than the next-best gauge product ($E_7 \times E_8$, 8.80%).

Holonomy Ranking

Rank	Holonomy	Mean Dev.
1	G_2	0.73%
2	$SU(4)$	1.46%
3	$SU(3)$	4.43%

G_2 achieves approximately 6x better agreement than Calabi-Yau ($SU(3)$, 4.43%).

Interpretation

Among 192,349 tested alternatives, the configuration ($b_2=21$, $b_3=77$) with $E_8 \times E_8$ gauge group and G_2 holonomy achieves the lowest mean deviation. Zero alternatives outperform it.

Literature uniqueness: Analysis of the NK-certified metric ([B], §6) compared against all ~65 known compact G_2 manifolds from the literature (Kovalev, CHNP, CGN, Joyce, Nordström). The point ($b_2=21$, $b_3=77$) is **UNIQUE**: no known diffeomorphic example exists. The nearest neighbor in (b_2,b_3) space is the CHNP grid point (9,9), at distance 7.6. The topological pair is realized by the Joyce-Karigiannis Z_2^3 orbifold route (S1 §8.4), extending the known catalogue; an explicit closed-form neck ansatz is established in [D].

Complete methodology: `STATISTICAL_EVIDENCE.md` in the framework repository (available on request).

Part IX: Observable Catalog

24. Structural Redundancy and Expression Counts

Each prediction admits multiple algebraically independent expressions that reduce to the same fraction. This multiplicity provides a measure of structural robustness: quantities arising from many paths through the topological invariants are less likely to represent numerical coincidence.

24.1 Classification Scheme

Classification	Expressions	Interpretation
CANONICAL	≥ 20	Maximally over-determined; emerges from algebraic web
ROBUST	10-19	Highly constrained; multiple independent derivations
SUPPORTED	5-9	Structural redundancy
DERIVED	2-4	Dual derivation minimum
SINGULAR	1	Unique path (possible coincidence)

24.2 Core 18 Type I Predictions with Expression Counts (subset of 33 certified)

#	Observable	Formula	Value	Exp.	Dev.	Expr.	Class
1	N_gen	Atiyah-Singer	3	3	0.00%	24+	CANONICAL
2	$\sin^2\theta_W$	$b_2/(b_3+\dim_{G_2})$	3/13	0.2312	0.20%	14	ROBUST
3	$\alpha_s(M_Z)$	$\sqrt{2}/12$	0.11785	0.11800	0.126%	9	SUPPORTED
4	λ_H	$\sqrt{17}/32$	0.1288	0.129	0.12%	4	DERIVED
5	α^{-1}	128+9+corr	137.033	137.036	0.002%	3	DERIVED
6	Q_Koide	$\dim_{G_2}/b_2$	2/3	0.6667	0.001%	20	CANONICAL
7	m_τ/m_e	$7+10\times 248+10\times 9977$	3477	3477.2	0.004%	3	DERIVED

#	Observable	Formula	Value	Exp.	Dev.	Expr.	Class
8	m_μ/m_e	$27^\wedge \varphi$	207.01	206.77	0.12%	2	DERIVED
9	m_s/m_d	$p_2^2 \times \text{Weyl}$	20	20.0	0.00%	14	ROBUST
10	m_b/m_t	$1/(2b_2)$	1/42	0.024	0.79%	21	CANONICAL
11	m_u/m_d	$(1+\dim_{E_6})/\text{PSL}_{27}$	79/168	0.47	0.05%	1	SINGULAR
12	δ_{CP}	$7 \times 14 + 99$	197°	$177^\circ \pm 20^\circ$	11.3% (1σ)	3	DERIVED
13	θ_{13}	π/b_2	8.57°	8.54°	0.37%	3	DERIVED
14	θ_{23}	$\arcsin((b_3 - p_2)/H^*)$	49.25°	49.3°	0.10%	2	DERIVED
15	θ_{12}	$\arctan(\sqrt{(\delta/\gamma)})$	33.40°	33.41°	0.03%	2	DERIVED
16	Ω_{DE}	$\ln(2) \times (b_2 + b_3)/10^8$	0.6861	0.6847	0.21%	2	DERIVED
17	n_s	$\zeta(11)/\zeta(5)$	0.9649	0.9649	0.004%	2	DERIVED
18	$\det(g)$	65/32	2.0313	-	-	8	SUPPORTED

Distribution: 3 CANONICAL (17%), 2 ROBUST (11%), 2 SUPPORTED (11%), 10 DERIVED (56%), 1 SINGULAR (6%).

24.3 Extended Certified Predictions (#19–#33, 15 entries)

#	Observable	Formula	Value	Exp.	Dev.	Expr.	Class
19	$\sin^2 \theta_{12}^{\text{PMNS}}$	$(1 + N_{\text{gen}})/\alpha_{\text{sum}}$	4/13	0.307	0.23%	28	CANONICAL
20	$\sin^2 \theta_{23}^{\text{PMNS}}$	$(D_{\text{bulk-Weyl}}/D_{\text{bulk}})$	6/11	0.546	0.10%	15	ROBUST
21	$\sin^2 \theta_{13}^{\text{PMNS}}$	$D_{\text{bulk}}/\dim_{E_{82}}$	11/496	0.022	0.81%	5	SUPPORTED
22	$\sin^2 \theta_{12}^{\text{CKM}}$	7/31	0.2258	0.225	0.36%	16	ROBUST
23	A_{Wolf}	$(\text{Weyl} + \dim_{E_6})/H^*$	83/99	0.836	0.29%	4	DERIVED
24	$\sin^2 \theta_{23}^{\text{CKM}}$	$\dim_{K_7/\text{PSL}_{27}}$	1/24	0.041	1.13%	3	DERIVED
25	m_H/m_t	8/11	0.7273	0.725	0.31%	19	ROBUST
26	m_H/m_W	81/52	1.5577	1.558	0.02%	1	SINGULAR
27	m_W/m_Z	$(2b_2 - \text{Weyl})/(2b_2)$ $= 37/42$	0.8810	0.8815	0.06%	8	SUPPORTED
28	m_μ/m_τ	5/84	0.0595	0.0595	0.04%	9	SUPPORTED

#	Observable	Formula	Value	Exp.	Dev.	Expr.	Class
29	$\Omega_{\text{DM}}/\Omega_{\text{b}}$	$(1+42)/\text{rank_E}_8$	43/8	5.375	0.00%	6	SUPPORTED
30	$\Omega_{\text{b}}/\Omega_{\text{m}}$	$\text{Weyl}/\det(\text{g_den})$	5/32	0.156	0.16%	7	SUPPORTED
31	$\Omega_{\Lambda}/\Omega_{\text{m}}$	$(\det_{\text{g_den}} - \dim_{\text{K}_7})/\text{D_bulk}$	25/11	2.27	0.12%	6	SUPPORTED
32	h	$(\text{PSL}_{27-1})/\dim_{\text{E}_8}$	167/248	0.674	0.09%	3	DERIVED
33	σ_8	$(\text{p}_2 + \det_{\text{g_den}})/(2\text{b}_2)$	34/42	0.811	0.18%	4	DERIVED

24.3b Relations #34–#35: BH Remnant Topological Predictions: Type IV (Pinčák et al. 2026 [42])

These are classified as **Type IV structural diagnostics** ($D_{22} = M_{\text{res}}$, $D_{20} = N_{\text{QNM}}$) in the 95-observable dataset, not as Type I. They are included here for completeness as they are topologically derived from the same G_2 structural constants.

#	Observable	Formula	Value	Status
34	BH remnant mass M_{res}	$\text{v_EW}^2/\text{M_Pl}$ (torsion VEV: $\tau_0 = \text{v_EW}$)	$\sim 125 \text{ GeV}^2 / \text{M_Pl}$	Type IV; topological, not yet Lean-certified
35	QNM families N_{QNM}	$\text{b}_3 + \text{Weyl} \times \dim(\text{K}_7)$ + p_2 (G_2 torsion modes)	98	Type IV; topological, not yet Lean-certified

These two predictions are topologically derived from G_2 torsion structure but have not yet been formally certified in Lean. M_{res} involves a physical scale identification ($\tau_0 = \text{v_EW}$); N_{QNM} is a purely structural count. No experimental comparison is currently available.

24.4 Illustrative Examples of Multiple Expressions

$\sin^2\theta_{\text{W}} = 3/13$ (14 independent expressions):

#	Expression	Evaluation
1	$N_{\text{gen}} / \alpha_{\text{sum}}$	3/13
2	$N_{\text{gen}} / (\text{p}_2 + \text{D_bulk})$	$3/(2+11) = 3/13$
3	$\text{b}_2 / (\text{b}_3 + \dim_{G_2})$	$21/91 = 3/13$
4	$\dim(\text{J}_3\text{O}) / (\dim_{\text{F}_4} + \det_{\text{g_num}})$	$27/117 = 3/13$

#	Expression	Evaluation
5	$(b_0 + \dim_G_2) / \det_g_num$	$15/65 = 3/13$
6	$(p_2 + b_0) / \alpha_sum$	$3/13$
7	$\dim_K_7 / (b_2 + \dim_K_7 + \dim_G_2)$	$7/42 \neq 3/13 \times$

(Expression 7 illustrates that not all combinations work; only those reducing to $3/13$ are valid.)

Q_Koide = **2/3** (20 independent expressions):

#	Expression	Evaluation
1	p_2 / N_gen	$2/3$
2	$\dim_G_2 / b_2$	$14/21 = 2/3$
3	$\dim_F_4 / \dim_E_6$	$52/78 = 2/3$
4	$\text{rank_}E_8 / (\text{Weyl} + \dim_K_7)$	$8/12 = 2/3$
5	$(\dim_G_2 - \text{rank_}E_8) / (\text{rank_}E_8 + 1)$	$6/9 = 2/3$

m_b/m_t = **1/42** (21 independent expressions):

#	Expression	Evaluation
1	$b_0 / (2b_2)$	$1/42$
2	$(b_0 + N_gen) / \text{PSL}(2,7)$	$4/168 = 1/42$
3	$p_2 / (\dim_K_7 + b_3)$	$2/84 = 1/42$
4	$N_gen / (\dim(J_3O) + H^*)$	$3/126 = 1/42$
5	$\dim_K_7 / (\dim_E_8 + \dim(J_3O) + \dim_K_7)$	$7/294 = 1/42$

The ratio $m_b/m_t = 1/42 = 1/(2b_2)$ illustrates structural redundancy: the bottom-to-top mass hierarchy equals the inverse of the structural constant $2b_2 = p_2 \times b_2$.

Note: The true Euler characteristic $\chi(K_7) = 0$ for G_2 manifolds (odd-dimensional). The constant 42 is the structural invariant $2b_2$.

24.5 The Algebraic Web

The topological constants satisfy interconnected identities:

Identity	Left side	Right side
Fiber-holonomy	$\dim(G_2) = 14$	$p_2 \times \dim(K_7) = 2 \times 7$
Gauge moduli	$b_2 = 21$	$N_gen \times \dim(K_7) = 3 \times 7$
Matter-holonomy	$b_3 + \dim(G_2) = 91$	$\dim(K_7) \times \alpha_sum = 7 \times 13$
Fano order	$\text{PSL}(2,7) = 168$	$\text{rank}(E_8) \times b_2 = 8 \times 21$
Fano order	$\text{PSL}(2,7) = 168$	$N_gen \times \text{fund}(E_7) = 3 \times 56$

Identity	Left side	Right side
Anomaly sum	$\alpha_sum = 13$	$rank(E_8) + Weyl = 8 + 5$

These relations form a closed algebraic system. The mod-7 structure ($\dim(K_7) = 7$ divides $\dim(G_2)$, b_2 , b_3 , $PSL(2,7)$) reflects the Fano plane underlying octonion multiplication.

24.6 Fibonacci-Lucas Embedding

The framework’s constants embed naturally into the Fibonacci (F_n) and Lucas (L_n) sequences:

n	F_n	Constant	Role
3	2	p_2	Pontryagin class
4	3	N_gen	Fermion generations
5	5	Weyl	Pentagonal symmetry
6	8	$rank(E_8)$	Cartan subalgebra
7	13	α^2_B sum	Structure coefficient
8	21	b_2	Second Betti number

This sequence propagates via the recurrence:

$$F_3 + F_4 = F_5 \quad \Rightarrow \quad p_2 + N_{gen} = Weyl$$

Lucas numbers also appear naturally:

L_n	Value	Role
L_4	7	$\dim(K_7)$
L_5	11	D_bulk
L_8	47	Scale bridge exponent

The Lucas identity $L_8 = F_7 + F_9 = 13 + 34$ decomposes as:

$$L_8 = \alpha^B_{sum} + d_{hidden} = 13 + 34 = 47$$

This structure reflects the icosahedral geometry underlying the McKay correspondence $E_8 \leftrightarrow 2I$, where icosahedral coordinates involve the golden ratio $\varphi = \lim(F_{n+1}/F_n)$.

Status: STRUCTURAL (mathematical pattern; physical significance unclear)

Appendix F: δ_{CP} : The 197° Prediction and the Compactification Correction

F.1 The Original Prediction

The framework's prediction $\delta_{\text{CP}} = 197^\circ = 7 \times 14 + 99 = \dim(K_7) \times \dim(G_2) + H^*$ matches NuFIT 5.2 (2022: $197^\circ \pm 24^\circ$) exactly.

NuFIT 6.0 (October 2024) shifted the central value to $177^\circ \pm 20^\circ$. The 197° prediction is at the 1σ boundary of this measurement. The NuFIT collaboration notes that δ_{CP} remains one of the least constrained oscillation parameters, with significant sensitivity to reactor flux model assumptions and the mass ordering.

NuFIT 6.1 update (2025). The subsequent release, incorporating the T2K+NOvA joint analysis, shifted the central value to $207^\circ +23/-20$ (NO w/o SK-atm): the 197° prediction now sits *inside* the 1σ band (deviation 4.8% from centre). The two most recent releases bracket the prediction on both sides; the discussion below, framed against NuFIT 6.0, is retained as the frozen-dataset baseline.

The canonical framework prediction remains $\delta_{\text{CP}} = 197^\circ$.

F.2 The Compactification Factor 62/69

PSLQ residual analysis (§7.6 of main paper) identifies a potential correction factor:

$$\frac{62}{69} = \frac{\dim(E_8)}{\dim(E_8) + 4 \dim(K_7)} = \frac{248}{248 + 28} = \frac{248}{276}$$

This factor has a clean structural interpretation: the ratio of gauge degrees of freedom ($\dim(E_8) = 248$) to total degrees of freedom ($248 \text{ gauge} + 4 \times 7 \text{ gravitational} = 276$).

Applied to the original prediction:

$$197 \times \frac{62}{69} = \frac{12214}{69} = 177.01^\circ$$

This matches NuFIT 6.0 to **0.008%**. The formula $12214/69$ uses only the framework's topological integers and the compactification ratio.

F.3 Why We Do Not Adopt It (Yet)

1. **Post-hoc identification:** The factor 62/69 was found AFTER the NuFIT 6.0 shift. Adopting it would be fitting to the latest central value: the opposite of prediction.
2. **197° remains within 1σ :** The value $197^\circ = 177^\circ + 20^\circ$ is AT the 1σ boundary. One-sigma deviations are expected $\sim 32\%$ of the time. This is not a falsification.
3. **Experimental instability:** δ_{CP} shifted 20° between NuFIT 5.2 and 6.0. The next update could shift back. DUNE will provide the definitive measurement with resolution of a few degrees to $\sim 15^\circ$ (2028–2040).

4. **Formula integrity:** The framework’s predictions are derived from topology, not fitted to experiment. The 197° formula (3 constants, 2 operations) is cleaner than the 177.01° formula (5 constants, 4 operations).

F.4 Contingency Plan

If DUNE confirms $\delta_{\text{CP}} \approx 177^\circ$ (central value within $\pm 5^\circ$ of 177°): $\rightarrow$ Adopt $12214/69 = 177.01^\circ$ as the primary prediction $\rightarrow$ Interpret $62/69$ as a compactification factor (gauge/total DOF ratio) $\rightarrow$ The original 197° becomes the “bare” topological value before compactification

If DUNE confirms $\delta_{\text{CP}} \approx 197^\circ$ (central value within $\pm 10^\circ$ of 197°): $\rightarrow$ Original prediction vindicated $\rightarrow$ The $62/69$ factor was a false lead (PSLQ artifact)

If DUNE gives a value inconsistent with both 177° and 197° : $\rightarrow$ Both formulas fail; the neutrino sector requires revision

F.5 Summary

Formula	Value	Status	Complexity
$7 \times 14 + 99$	197°	PRIMARY (original prediction)	3 constants, 2 ops
$12214/69$	177.01°	Contingent (pending DUNE)	5 constants, 4 ops
Experimental	$177^\circ \pm 20^\circ$	NuFIT 6.0 (1σ band: $[157^\circ, 197^\circ]$); NuFIT 6.1: $207^\circ + 23/-20$	n/a

Author’s Related Works

- **[A]** B. de La Fournière, “An Explicit Approximate G_2 Metric on a Compact 7-Manifold with Certified Torsion-Free Completion,” Zenodo [10.5281/zenodo.19892350](https://zenodo.org/record/105281/files/19892350) (2026).
- **[B]** B. de La Fournière, “Spectral Geometry of the G_2 -GIFT Manifold: Betti Numbers, KK Spectrum, and Spectral Invariants,” Zenodo [10.5281/zenodo.19893371](https://zenodo.org/record/105281/files/19893371) (2026).
- **[D]** B. de La Fournière, “An Explicit Closed-Form G_2 Ansatz on a K3-Coassociative Neck with Hyperkähler Rotation and Picard-Lefschetz Wirtinger Certificate,” Zenodo [10.5281/zenodo.20039066](https://zenodo.org/record/105281/files/20039066) (2026).

References

- [1] Particle Data Group, “Review of Particle Physics,” *Phys. Rev. D* **110**, 030001 (2024).
- [2] S. Weinberg, “Implications of dynamical symmetry breaking,” *Phys. Rev. D* **13**, 974 (1976).
- [3] Planck Collaboration, “Planck 2018 results. VI. Cosmological parameters,” *Astron. Astrophys.* **641**, A6 (2020).
- [4] A. G. Riess et al., “A comprehensive measurement of the local value of the Hubble constant,” *ApJL* **934**, L7 (2022).
- [5] C. D. Froggatt, H. B. Nielsen, “Hierarchy of quark masses, Cabibbo angles and CP violation,” *Nucl. Phys. B* **147**, 277 (1979).
- [6] Y. Koide, “Fermion-boson two-body model of quarks and leptons and Cabibbo mixing,” *Lett. Nuovo Cim.* **34**, 201 (1982).
- [7] C. Furey, *Standard Model Physics from an Algebra?*, PhD thesis, University of Waterloo (2015).
- [8] N. Furey, M. J. Hughes, “One generation of standard model Weyl representations as a single copy of $\mathbb{R} \otimes \mathbb{C} \otimes \mathbb{H} \otimes \mathbb{O}$,” *Phys. Lett. B* **831**, 137186 (2022).
- [9] R. Wilson, “ E_8 and Standard Model plus gravity,” arXiv:2404.18938 [hep-th] (2024).
- [10] T. P. Singh et al., “An $E_8 \otimes E_8$ unification of the Standard Model with pre-gravitation,” arXiv:2206.06911v3 (2024).
- [11] B. S. Acharya, S. Gukov, “M-theory and singularities of exceptional holonomy manifolds,” *Phys. Rep.* **392**, 121 (2004).
- [12] L. Foscolo et al., “Complete non-compact G_2 -manifolds from asymptotically conical Calabi–Yau 3-folds,” *Duke Math. J.* **170**, 3 (2021).
- [13] A. Kovalev, “Coassociative K3 fibrations of compact G_2 -manifolds,” arXiv:math/0511150 (2005).
- [14] M. Haskins, J. Nordström, “Extra-twisted connected sum G_2 -manifolds,” arXiv:1809.09083 (2022); *Ann. Glob. Anal. Geom.* **64**, art. 2 (2023).
- [15] G. Bera, “Associative submanifolds in twisted connected sum G_2 -manifolds,” arXiv:2209.00156 (2022).
- [16] D. Crowley, S. Goette, J. Nordström, “An analytic invariant of G_2 manifolds,” *Invent. Math.* **239**(3), 865–907 (2025).
- [17] T. Dray, C. A. Manogue, *The Geometry of the Octonions*, World Scientific (2015).
- [18] J. F. Adams, *Lectures on Exceptional Lie Groups*, University of Chicago Press (1996).
- [19] D. J. Gross et al., “Heterotic string theory,” *Nucl. Phys. B* **256**, 253 (1985).
- [20] D. D. Joyce, *Compact Manifolds with Special Holonomy*, Oxford University Press (2000).

- [21] A. Kovalev, “Twisted connected sums and special Riemannian holonomy,” *J. Reine Angew. Math.* **565**, 125 (2003).
- [22] A. Corti et al., “ G_2 -manifolds and associative submanifolds via semi-Fano 3-folds,” *Duke Math. J.* **164**, 1971 (2015).
- [23] R. Harvey, H. B. Lawson, “Calibrated geometries,” *Acta Math.* **148**, 47 (1982).
- [24] B. S. Acharya, “M-theory, G_2 -manifolds and four-dimensional physics,” *Class. Quant. Grav.* **19**, 5619 (2002).
- [25] B. S. Acharya, E. Witten, “Chiral fermions from manifolds of G_2 holonomy,” arXiv:hep-th/0109152 (2001).
- [26] T2K and NOvA Collaborations, “Joint analysis of long-baseline neutrino oscillations,” *Nature* **646**, 818–824 (2025), doi:10.1038/s41586-025-09599-3.
- [27] I. Esteban et al. (NuFIT), “NuFit-6.0: three-flavour global analysis of neutrino oscillation experiments,” *JHEP* **12** (2024) 216, arXiv:2410.05380; NuFIT 6.1 web release, <https://www.nu-fit.org> (2025).
- [28] L. de Moura, S. Ullrich, “The Lean 4 theorem prover and programming language,” in *CADE 28*, p. 625 (2021).
- [29] mathlib Community, *mathlib4*, <https://github.com/leanprover-community/mathlib4>.
- [30] DUNE Collaboration, “Long-baseline neutrino facility (LBNF) and DUNE conceptual design report,” FERMILAB-TM-2696 (2020).
- [31] DUNE Collaboration, “Deep Underground Neutrino Experiment (DUNE) far detector technical design report,” arXiv:2103.04797 (2021).
- [32] E. Witten, “Strong coupling expansion of Calabi-Yau compactification,” *Nucl. Phys. B* **471**, 135 (1996).
- [33] B. S. Acharya et al., “M-theory solution to the hierarchy problem,” *Phys. Rev. D* **76**, 126010 (2007).
- [34] M. Atiyah, E. Witten, “M-theory dynamics on a manifold of G_2 -holonomy,” *Adv. Theor. Math. Phys.* **6**, 1 (2002).
- [35] G. Kane, *String Theory and the Real World* (2017).
- [36] J. Distler, S. Garibaldi, “There is no ‘Theory of Everything’ inside E_8 ,” *Commun. Math. Phys.* **298**, 419 (2010).
- [37] J. C. Baez, “Octonions and the Standard Model,” <https://math.ucr.edu/home/baez/standard/> (2020–2025).
- [38] Springer Nature, “Artificial intelligence (AI) policy,” <https://www.springernature.com/gp/policies> (2024).

- [39] J. A. Wheeler, “Information, physics, quantum: the search for links,” in *Complexity, Entropy, and the Physics of Information*, W. H. Zurek (ed.), Addison-Wesley (1990), pp. 3–28.
- [40] J. Worrall, “Structural realism: the best of both worlds?,” *Dialectica* **43**, 99–124 (1989).
- [41] J. Ladyman, D. Ross, *Every Thing Must Go: Metaphysics Naturalized*, Oxford University Press (2007).
- [42] R. Pinčák, A. Pigazzini, M. Pudlák, E. Bartoš, “Geometric origin of a stable black hole remnant from torsion in G_2 -manifold geometry,” *Gen. Rel. Grav.* **58**, 29 (2026), [doi:10.1007/s10714-026-03528-z](https://doi.org/10.1007/s10714-026-03528-z).
- [43] D. Joyce, S. Karigiannis, “A new construction of compact torsion-free G_2 -manifolds by gluing families of Eguchi–Hanson spaces,” *J. Differential Geom.* (2021), arXiv:1707.09325 (2017).
- [44] S. Mukai, “Finite groups of automorphisms of K3 surfaces and the Mathieu group,” *Invent. Math.* **94**, 183–221 (1988).
- [45] A. Garbagnati, A. Sarti, “Symplectic automorphisms of prime order on K3 surfaces,” *J. Algebra* **318**, 323–350 (2007), arXiv:math/0603742.
- [46] LEP SUSY Working Group (ALEPH, DELPHI, L3, OPAL), “Combined LEP chargino results, up to 208 GeV for large m_0 ,” LEPSUSYWG/01-03.1 (2001), https://lepsusy.web.cern.ch/lepsusy/www/inos_moriond01/charginos_pub.html.
- [47] Fermilab, “DUNE first-beam target of 2031: LBNF/DUNE-US milestone communication” (2026-05-07), <https://news.fnal.gov/>.
- [48] J. Chen, H. Hong, “Intermediate curvature and splitting theorem,” arXiv:2604.26529 (2026).
- [49] M. Tristram et al., “Cosmological parameters derived from the final Planck data release (PR4),” *Astron. Astrophys.* **682**, A37 (2024).
- [50] E. Tiesinga, P. J. Mohr, D. B. Newell, B. N. Taylor, “CODATA recommended values of the fundamental physical constants: 2022,” *Rev. Mod. Phys.* **97**, 025002 (2025), <https://physics.nist.gov/cuu/Constants>.
- [51] V. V. Nikulin, “Integer symmetric bilinear forms and some of their geometric applications,” *Izv. Akad. Nauk SSSR Ser. Mat.* **43** (1979); English transl. *Math. USSR Izv.* **14** (1980).